

Savitzky-Golay split command-lag search

rotor median publish period [s]: 0.00499773025513
gimbal median publish period [s]: 0.00499868392944
initial lags [s]: {'rotor': 0.004997730255126953, 'gimbal': 0.004998683929443359}
smooth half-width multipliers: [4.0, 2.0, 1.0, 0.5]

smooth width=4: rotor=0.2s gimbal=0.0238885084s cost=861.277636043 lag_gradient={'rotor':
-2.2216414104490845, 'gimbal': -0.08024006570449116} FD=[{'direction': 'negative_gradient',
'scheme': 'backward', 'step': 6.0554544523933395e-06, 'analytic_l2_norm': 8.051642044201921,
'finite_difference_l2_norm': 8.051639241915753, 'difference_l2_norm':
0.0008081679241969332, 'relative_error': 0.00010037305679515449}, {'direction':
'rotor_delay_seconds', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 8.154538321652208, 'finite_difference_l2_norm': 8.154537543539771,
'difference_l2_norm': 0.000809124153041179, 'relative_error': 9.922378448976871e-05},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 18.525084059580227, 'finite_difference_l2_norm': 18.5252698979772,
'difference_l2_norm': 0.01451862087074637, 'relative_error': 0.0007837089265391473}]
smooth width=2: rotor=0.2s gimbal=0.0235268278s cost=861.28820238 lag_gradient={'rotor':
-1.028781066468, 'gimbal': 0.5832777762136838} FD=[{'direction': 'negative_gradient',
'scheme': 'backward', 'step': 6.0554544523933395e-06, 'analytic_l2_norm': 14.07528166578196,
'finite_difference_l2_norm': 14.075204051586, 'difference_l2_norm': 0.002998293608053289,
'relative_error': 0.0002130183735748863}, {'direction': 'rotor_delay_seconds', 'scheme':
'backward', 'step': 6.0554544523933395e-06, 'analytic_l2_norm': 9.036632353144997,
'finite_difference_l2_norm': 9.036637952468691, 'difference_l2_norm': 0.0031338856131501013,
'relative_error': 0.00034679773933999036}, {'direction': 'gimbal_delay_seconds', 'scheme':
'central', 'step': 6.0554544523933395e-06, 'analytic_l2_norm': 20.635698062545902,
'finite_difference_l2_norm': 20.593411809222495, 'difference_l2_norm': 0.6892440901763746,
'relative_error': 0.03340057060765794}]
smooth width=1: rotor=0.2s gimbal=0.0224374665s cost=861.285739101 lag_gradient={'rotor':
-1.6664661617524654, 'gimbal': 0.14331223409915395} FD=[{'direction': 'negative_gradient',
'scheme': 'backward', 'step': 6.0554544523933395e-06, 'analytic_l2_norm':
11.273491065941244, 'finite_difference_l2_norm': 11.274227682575097, 'difference_l2_norm':
0.009942758976510374, 'relative_error': 0.0008819015595966208}, {'direction':
'rotor_delay_seconds', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 10.873225500610713, 'finite_difference_l2_norm': 10.874074535564949,
'difference_l2_norm': 0.010014268832355603, 'relative_error': 0.0009209306777880499},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 24.75571976940552, 'finite_difference_l2_norm': 24.739459686847233,
'difference_l2_norm': 0.39839270027639107, 'relative_error': 0.0160929556477185}]
smooth width=0.5: rotor=0.2s gimbal=0.0214927525s cost=861.276787174 lag_gradient={'rotor':
-7.234182630924998, 'gimbal': -1.6819457531466933} FD=[{'direction': 'negative_gradient',
'scheme': 'backward', 'step': 6.0554544523933395e-06, 'analytic_l2_norm':
15.462671214689621, 'finite_difference_l2_norm': 15.466799864022105, 'difference_l2_norm':
0.030446235070347438, 'relative_error': 0.001968489625392357}, {'direction':
'rotor_delay_seconds', 'scheme': 'backward', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 13.568240323557122, 'finite_difference_l2_norm': 13.572577472872982,
'difference_l2_norm': 0.031631366714371625, 'relative_error': 0.002330534990689284},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 37.969663910736465, 'finite_difference_l2_norm': 37.99916226656394,
'difference_l2_norm': 1.7609011820560456, 'relative_error': 0.046340526396432956}]

strict profile iterations: 1

iteration 1: best=(0.190004539397, 0.021492752532) cost=861.268003963
rotor_range=(0.185006809142, 0.2) gimbal_range=(0.016494068602, 0.026491436461)

```
expansions=[{'best_before_expansion': {'rotor_delay_seconds': 0.195002269652,  
'gimbal_delay_seconds': 0.021492752532, 'cost': 861.2788360842837}, 'added': [{'axis':  
'rotor', 'delay_seconds': 0.190004539397}]}, {'best_before_expansion':  
{'rotor_delay_seconds': 0.190004539397, 'gimbal_delay_seconds': 0.021492752532, 'cost':  
861.2680039625353}, 'added': [{'axis': 'rotor', 'delay_seconds': 0.185006809142}]]
```

```
selected rotor lag 0.190004539397 s (38.0182 publish periods)  
selected gimbal lag 0.021492752532 s (4.29968 publish periods)  
selected objective cost 861.267977611
```